Kothapalli Madanmohan Reddy

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Career Objective

To leverage my expertise in Artificial Intelligence and Data Science to develop cutting-edge machine learning models and innovative solutions, contributing to business growth and societal progress. I aim to enhance operational efficiency and customer experience by deploying scalable AI-driven applications, ensuring continuous learning and impactful outcomes.

Internship Experience

Naresh I Technologies Aug 2024 – Jan 2025
Data Science Intern *Hyderabad, India*

- **Project Name:** Real Estate Price Prediction
- **Tech Stack:** Python, Pandas, NumPy, Scikit-learn, XGBoost, Streamlit, Git
- **Project Description:** Built a regression-based ML model to predict real estate prices with 89% R^2 accuracy.
- **Roles & Responsibilities:**
  - Conducted feature engineering and preprocessing to improve model robustness.
  - Developed a Streamlit app for real-time prediction and data visualization.
  - Performed hyperparameter optimization using GridSearchCV for enhanced performance.
  - Utilized Git for version control and collaborative development.
  - Delivered detailed documentation and presentation of the model's architecture and results.

AICTE Jan 2024 – Mar 2024
AI Project Intern *Remote*

- **Project Name:** Mental Fitness Tracker
- **Tech Stack:** Python, NLTK, Flask, Streamlit, Jupyter Notebook
- **Project Description:** Designed a sentiment analysis tool to evaluate mental health using labeled emotional datasets.
- **Roles & Responsibilities:**
  - Preprocessed textual data using NLTK for tokenization, lemmatization, and stopword removal.
  - Achieved 90% model accuracy with TF-IDF and logistic regression.
  - Built a Streamlit app integrated with Flask for real-time analysis and visualization.
  - Presented findings through Jupyter Notebook to enhance team insights.
  - Enhanced data pipelines to ensure seamless integration and scalability.

Education

B.Tech in Artificial Intelligence and Data Science Aug 2020 – Apr 2024
Nehru Institute of Engineering and Technology *Coimbatore, India — CGPA: 8.4*

Technical Skills

- **Programming Languages:** Python, SQL
- **Data Analysis & Manipulation:** Pandas, NumPy
- **Data Visualization:** Seaborn, Matplotlib, Plotly, Tableau, Power BI
- **Machine Learning:** Scikit-learn, XGBoost, Random Forest, Gradient Boosting, Cross-Validation, Hyperparameter Tuning, Predictive Modeling
- **Deep Learning:** TensorFlow, Keras, PyTorch
- **NLP & Generative AI:** Hugging Face Transformers, NLTK, OpenAI APIs, LLMs
- **Model Deployment & MLOps:** Streamlit, Flask, MLflow, Git, CI/CD (Basics)
- **Cloud Platforms:** AWS (Basic), Google Cloud Platform (Basic)
- **Collaboration Tools:** GitHub, Agile Methodologies

Projects

Traffic Intelligence: Predictive Traffic Management *Python, Flask, ARIMA*

- Developed a time series forecasting solution to predict traffic patterns and reduce congestion.
- Implemented ARIMA model and seasonal decomposition to detect peak trends and anomalies.
- Built a Flask dashboard to visualize traffic forecasts for city planners.
- Reduced error by 18% through optimal parameter tuning.

Movie Recommendation System *Python, Scikit-learn, Pandas, TF-IDF, Cosine Similarity*

- Implemented a hybrid recommendation engine combining collaborative and content-based filtering techniques.
- Boosted recommendation accuracy by 20% and user engagement by 25% using cross-validation and ML metrics.
- Deployed the model with real-time predictions for enhanced user experience.

Generative AI Text Summarization *Python, Hugging Face Transformers, Flask, Streamlit*

- Developed a generative AI-based text summarization application for efficient document analysis.
- Integrated pre-trained transformer models (BART, T5) for abstractive summarization tasks.
- Deployed a user-friendly Streamlit web interface for input, summary generation, and visualization.
- Improved summarization accuracy by fine-tuning the transformer models on domain-specific datasets.

Publications

- **SmartAgro: Enhancing Crop Management with Agribot IoT System** — Published in *IRO Journal of IoT in Social, Mobile, Analytics, and Cloud* (April 2024).
- Presented at the International Conference on Electronics, Communication, and Advanced Processing Systems (ICECAPS'24).
- Explored cutting-edge IoT and AI-driven innovations for precision agriculture, significantly improving farming efficiency and sustainability.
- Publication Link

Certificates

- Python Skill Certificate - HackerRank — Certificate Link
- Data Analytics and Visualization Job Simulation - Accenture — Certificate Link
- ChatGPT Course - GUVI Geek Networks — Certificate Link
- Data Science Course - Teachnook — Certificate Link

Extracurricular Activities

- Volunteering: Actively mentoring junior students in AI and Data Science to help develop their skills.
- Hobbies: Passionate about exploring new technologies, attending tech events, and participating in hackathons.

Achievements

- Achieved an 18% reduction in traffic congestion through predictive analytics in traffic management projects.
- Published research on IoT and AI for precision agriculture, contributing to sustainable farming innovations.
- Reduced consultation time by 30% in mental health analysis using NLP-driven solutions.

Declaration

I hereby declare that all the information provided above is true to the best of my knowledge and belief.

Place: Bengaluru, India **Kothapalli Madanmohan Reddy**